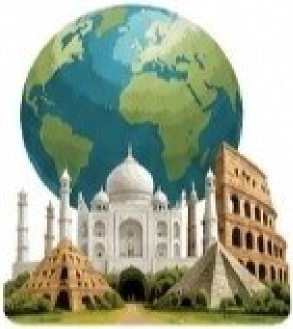

PROJECT DESIGN PHASE
SOLUTION ARCHITECTURE

Date	25 June 2026
Team ID	SWTID-2026-2982
Project Name	Heritage Treasures: Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

SOLUTION ARCHITECTURE

The proposed solution is an interactive Tableau-based dashboard that transforms UNESCO World Heritage Site data into meaningful visual insights. The architecture follows a simple data analytics workflow where raw data is collected, processed, analyzed, visualized, and presented to end users.

OBJECTIVES

- Provide a centralized platform to analyze UNESCO World Heritage Sites.
- Convert raw heritage data into interactive dashboards.
- Enable users to compare countries, regions, and heritage categories.
- Support researchers, students, tourists, and policy makers with visual insights.
- Help identify endangered heritage sites and historical trends.

MAIN COMPONENTS

1. **DATA SOURCE**
   - UNESCO World Heritage Dataset (CSV/Excel)
2. **DATA PROCESSING**
   - Data Cleaning
   - Remove Duplicates
   - Handle Missing Values
   - Data Formatting
3. **ANALYTICS LAYER**
   - Country-wise Analysis
   - Region-wise Analysis
   - Heritage Category Analysis
   - Endangered Site Analysis
   - Year-wise Trends
4. **VISUALIZATION LAYER**
   - Interactive Tableau Dashboard
   - Maps
   - KPI Cards
   - Charts
   - Filters
   - Story Navigation
5. **USERS**
   - Students
   - Researchers
   - Tourists
   - UNESCO Officials
   - Government Agencies

SOLUTION ARCHITECTURE FLOW

UNESCO Dataset (CSV / Excel) → Raw data collected from UNESCO Open Data Portal and other reliable sources.

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Data Collection & Import → Import dataset into the system for further processing.

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Data Cleaning & Pre-processing → Remove null values, duplicates, standardize formats, and handle missing values.

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Tableau Data Connection → Connect the cleaned data with Tableau for analysis and visualization.

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Data Analysis & Calculations →
- Category Analysis
- Country Analysis
- Regional Analysis
- Danger Status Analysis
- Trend Analysis

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Interactive Tableau Dashboard →
- KPI Cards
- World Map
- Bar Charts
- Pie Charts
- Line Charts
- Filters & Navigation

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Story Dashboard → Create storytelling dashboards to present insights effectively.

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End Users (Researchers · Tourists · Students · UNESCO · Government) → Users explore insights, make data-driven decisions and download reports.

TECHNOLOGY STACK

Tableau (Public/Desktop) → Excel / CSV (Data Source) → Data Processing & Cleaning → Data Modeling & Calculations → Dashboard Publishing (Tableau Public)

KEY BENEFITS

- Centralized analysis of global heritage data
- Interactive dashboards for better decision-making
- Compare countries, regions & categories easily
- Identify endangered sites & historical trends
- Export reports and share insights effectively

"Preserve Our Heritage, Understand Our Past, Shape Our Future."

Heritage Treasures Dashboard